eROSITA × DESI × AION-flow — Experiment Pipeline

Living document — keep in sync with the actual pipeline. Last updated: 2026-07-17. Render to PDF with `python docs/render_pipeline_pdf.py` (writes `docs/pipeline.pdf`).

1. Goal

Use a frozen astronomical foundation model (AION-1 base) as an information probe: predict eROSITA X-ray properties (broad-band flux, luminosity, hardness) and host stellar mass from non-X-ray DESI modalities (optical spectra, WISE, redshift, Legacy imaging) with a probabilistic normalizing-flow head. Report likelihood gain over a prior, not just point R^2. This repo extends the PAI26 paper with (a) a cleaned crossmatch and (b) uncertainty-aware, per-target training.

2. Compute

- **FASRC** (Harvard SLURM). Account **`finkbeiner_lab`** (confirmed working 2026-07-21); smokes on `gpu_test`, real runs on `gpu`. `siag_gpu` is *not* visible to us yet — it needs `siag_lab` group membership (Coldfront request pending).
- Env: `module load Miniforge3/26.1.0-fasrc01` + conda env + **cu124 torch wheel** + `pip install -e . polymathic-aion zuko safetensors wandb`. Built at `$AIONFLOW_ROOT/envs/aionflow` (torch 2.6.0+cu124).
- **Workflow:** develop locally → git push → `git pull` on the cluster; the SSH ControlMaster socket `~/.ssh/cm-fasrc` is used only to submit jobs and move data. All real work runs in `sbatch` jobs on compute nodes (login-node compute gets killed by arbiter2).
- **Tracking: Weights & Biases** (`--wandb`, project `erosita-desi-aion-flow`). Logs config, per-step batch NLL, per-epoch train/val NLL, and the final per-combination test metrics table. Auto-falls back to `offline` if a compute node lacks outbound internet (`wandb sync` later); a tracking failure never kills a job.

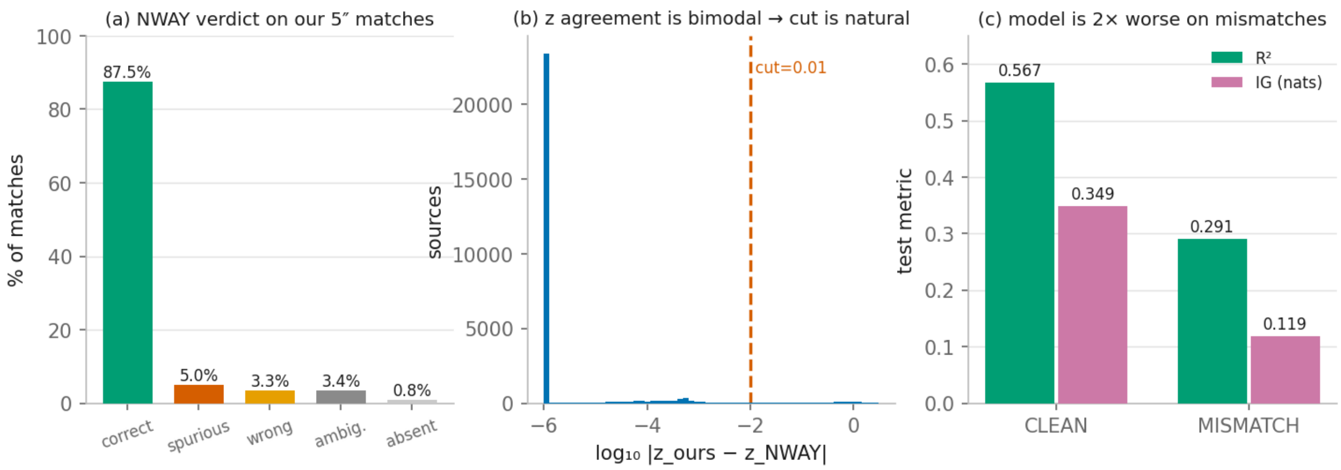
3. Data sources & download

Modality	Source	Notes
X-ray catalog	SRG/eROSITA-DE **DR1 / eRASS1** (Merloni+2024)	broad band `ML_FLUX_1` + sub-bands P1–P4 rates/fluxes **with asymmetric errors** (LOWERR/UPERR). Delivered as the eROSITA×DESI match CSVs.
Optical spectra	**DESI DR1** coadd spectra	per-healpix `coadd-{survey}-{program}-{healpix}.fits`, trimmed to matched targetids, compiled to `erosita_spectra_merged_32k.hdf5`.
Imaging	**Legacy Survey DR10** griz cutouts	160×160 px `fits_pool/<targetid>.fits` via legacysurvey.org cutout service (~11 GB).
Redshift / galaxy props	DESI DR1 VAC	z, `logmstar` (photometric), emission lines, `spectype` (QSO/GALAXY/STAR).
Foundation model	`polymathic-ai/aion-base` (HuggingFace, ~95 MB)	frozen tokenizer + backbone.

All inputs are local; only `aion-base` is internet-bound. "Adding sources our match missed" would require new DESI DR1 spectrum downloads (deferred).

4. Crossmatch & cleaning

- **Original match (existing):** naive **5″ nearest-neighbor** (`crossmatch_radec.py`, astropy) between DESI target RA/Dec and the eROSITA catalog → 33,938 pairs → 30,441 unique DESI targets → **28,646 with a retrieved spectrum**.
- **Quality problems found:** separations pile up near the 5″ cut (median 3.08″, 52% >3″); **14% of eROSITA sources match >1 DESI target** (26% of pairs in multi-match groups) → the same X-ray label duplicated onto several spectra.
- **Cleaning (NWAY):** validate against **Salvato et al. 2025 (A&A 704, A344)** — the eRASS1 Bayesian NWAY counterpart catalog — via VizieR `J/A+A/704/A344/dr1ls10` (join on `IAUName`; compare their counterpart's `zsp` to our matched DESI `z`). Per-`targetid` class: **87.5% correct, 4.6% spurious, 3.3% wrong, 3.4% ambiguous, 0.8% absent** → `keep = correct` filter (`data/match_quality.csv`).
- **Sanity checks (cleaning is sound, 3 independent angles):** (a) `|z_ours − z_NWAY|` is **bimodal** — 88.7% <1e-4 (same object), 3.0% >0.1 (different), only 0.6% in the 0.01–0.1 valley → the z-cut is natural, not tuned; (b) mismatches show larger match separation (3.19″ vs 3.06″) but only weakly — separation can't discriminate at eROSITA's ~arcsec position errors, which is *why* prior-based NWAY is needed; (c) the trained model is **2× worse on mismatches** (R^2 0.29 vs 0.57) — independent confirmation the flags localize real failures.
- **Pipeline order (IMPORTANT):** clean → **dedup to the NWAY counterpart per targetid** → **re-derive the train/val/test split on the cleaned sample** (targetid-grouped 80/10/10, seed 42, leakage-safe) → stage. Splits must be redone *after* cleaning.



5. Targets & uncertainties

Four separate 1-D flow runs (`data/targets_extra.csv` carries the derived targets + errors, keyed by `targetid`):

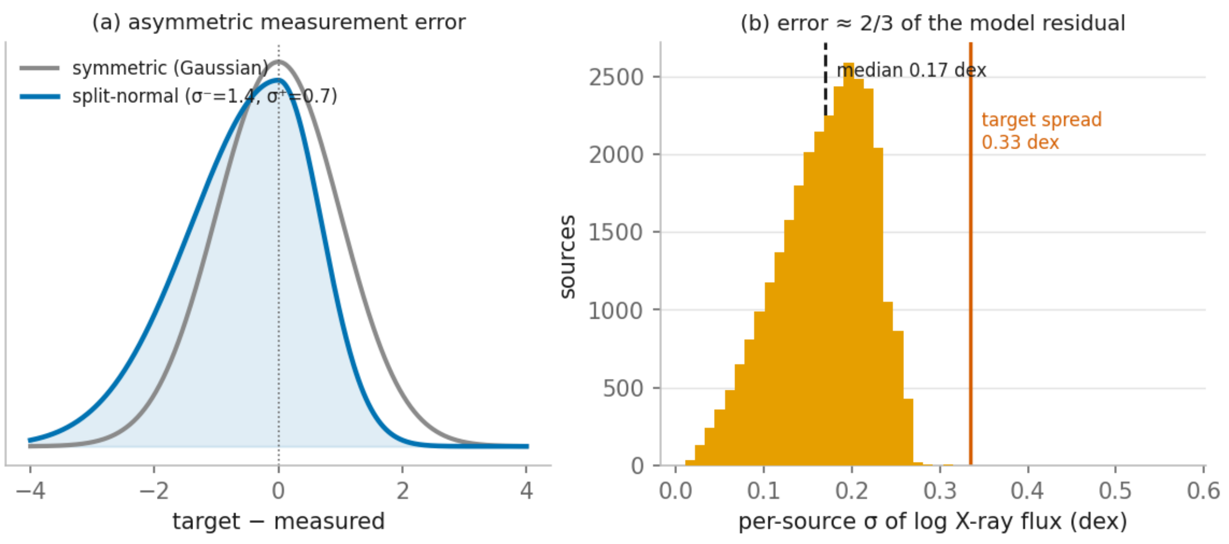
Target	Source	Error model
`log_ml_flux_1`	`log10(ML_FLUX_1)`	**split-normal** (two-piece Gaussian) from LOWERR/UPERR: `sig_hi=log10(1+UPERR/F)`, `sig_lo=-log10(1-LOWERR/F)` (capped). Median ≈0.17 dex; **98% asymmetric, low side heavier**.
`log_lx`	`ML_FLUX_1` + z (Planck18)	same dex σ as flux.
`logmstar`	DESI VAC photometric mass	**spectype systematic floor**: 0.20 dex (GALAXY) / 0.30 dex (QSO). No per-source posterior σ available (PROVABGS covers only 0.4% of this AGN sample). Caveat: 82% of sample is QSO → mass AGN-contaminated; evaluate split by `spectype`.

`hr32` (hardness)	`(P3−P2)/(P3+P2)` band rates	modeled in **arctanh space** `u=arctanh(HR)`, `sig_u=sig_hr/(1−HR²)`. Only ~17% have S/N>2 (σ≈signal) → **IG-primary**, R^2 secondary. HR43/Γ dropped.

Targets & error treatment (four separate V1 runs, all --error-mode convolve)

target	source	error model	mode	trainable
log_ml_flux_1	eROSITA ML_FLUX_1	split-normal (LO/HI), ~0.17 dex	convolve	full sample ✓
log_lx	flux + z (Planck18)	same dex σ as flux	convolve	full (z-dominated)
logmstar	DESI photometric mass	spectype floor 0.2/0.3 dex	convolve	clean for GALAXY; 82% QSO
hr32_u	eROSITA band rates (arctanh)	propagated σ_HR	convolve	IG-primary; ~17% S/N>2

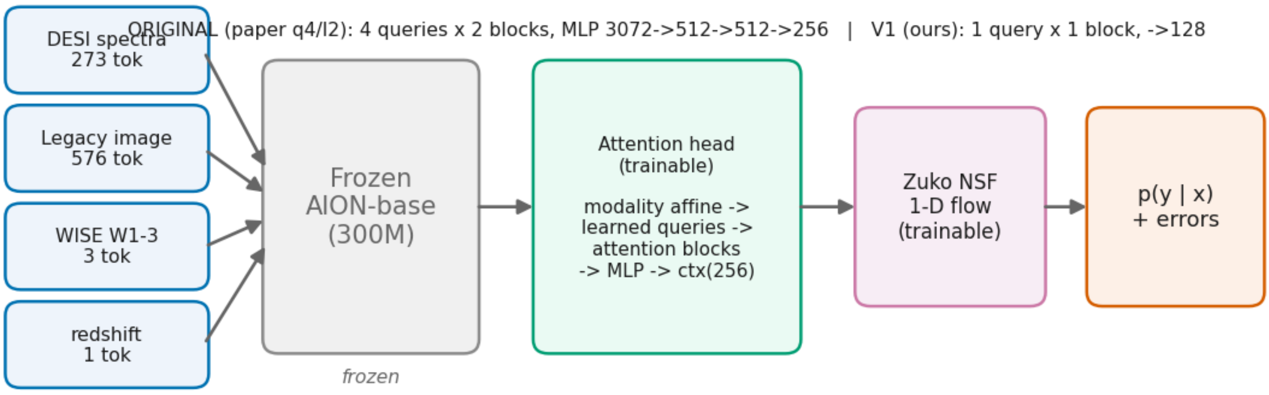
How errors enter the flow (split-normal convolution likelihood): `NLL_i = −log ∫ p_flow(t | x_i) · K(y_i | t; sig_lo_i, sig_hi_i) dt`, a fixed 1-D quadrature (~21–41 nodes over ±4σ), `error_mode ∈ {none, inject, convolve}` (default **convolve** = deconvolves; `inject` broadens; `none` = paper behavior).



Measurement error is large relative to signal: for flux, σ≈0.17 dex is ~2/3 of the model's residual variance → the model sits near the measurement-noise floor, so error-aware training + a reported error floor matter.

Deferred error refinement: X-ray sub-band non-detections are **upper limits (censored)**, not missing/large-error; proper handling via a censored likelihood (ref 2022A&A...661A...3S). First pass uses split-normal σ on measured values.

6. Model architecture

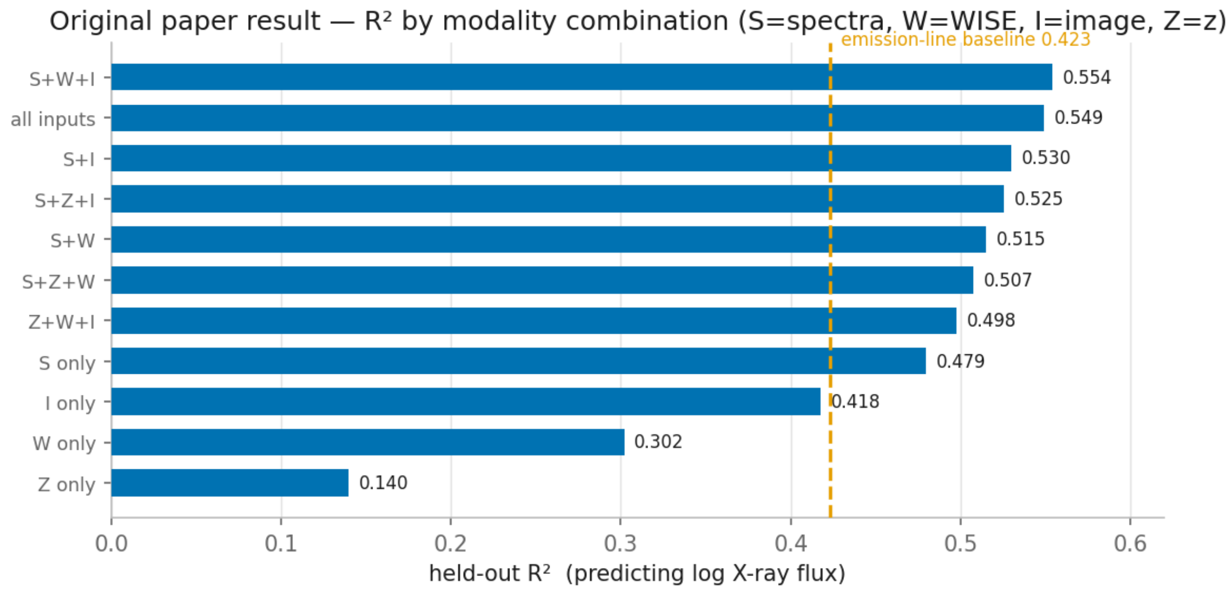


Per-batch a modality combo is sampled; AION frozen; KDE(target) prior anchors information gain.

Frozen **AION-base** tokenizer + backbone → per-modality token sequences (`spectra`=273, `image`=576, `WISE`=3, `z`=1; missing modalities omitted). Trainable head + flow:

- **V1 minimal head** (`AIONAttentionContext`, parameterized): per-modality affine calibration → **1 learned query + 1 attention block (8 heads)** cross/self-attention → 768 → MLP (→128) → 256-d context. (~7.8M params vs the paper's 4-query/3072-d ~16.8M.)
- **Flow:** Zuko conditional **NSF**, 1-D target, 8 transforms, context 256; standardized target; **KDE (Scott) prior** as the IG reference.
- Per-batch a modality combo is sampled; AION frozen; AdamW; checkpoint best all-inputs val NLL.
- **V2 (later):** replace frozen AION with a self-trained attention encoder over the raw spectrum.

Original paper result (q4/l2, epoch-13) — R^2 by modality combination; full spectra beat the emission-line baseline, all-modality is best:



7. Training & evaluation

- Four separate V1 runs (`--target {log_ml_flux_1,log_lx,logmstar,hr32}`), `--clean` (NWAY keep), `--error-mode convolve`, on `gpu`; smoke on `gpu_test` with a `--limit` subset.
- **Metrics:** primary **IG / exp(IG)** vs KDE prior (esp. HR); secondary R^2 + **reduced-χ^2** using per-source σ + the measurement-error floor; **clean-vs-full delta**; per-`spectype` split for logmstar/HR.
- Every run is mirrored to **W&B** (`--wandb`), so training curves are watchable live and runs are comparable across targets/error-modes via the run tags (`<target>`, `<error_mode>`, `V1`).

8. Key findings so far

- Naive 5″ match is **~8% wrong** per NWAY; the paper's q4/l2 model is **~2× worse on the mismatches (R^2 0.29 vs 0.57)**; dropping them lifts test R^2 from the published **0.549** → **0.567** (no retraining).
- X-ray targets are measurement-noise-dominated (flux σ≈0.17 dex; HR σ≈its own spread) → uncertainty handling is essential.

9. Status

- ■ Repo cloned; V1 head coded+tested; FASRC scaffolding + **verify-before-push staging scripts** (`scripts/fasrc_preflight.sh`, `scripts/fasrc_stage_data.sh`).
- ■ Sidecars built: `data/match_quality.csv` (clean filter), `data/targets_extra.csv` (targets + split-normal/arctanh errors).
- ■ **Error-aware flow** implemented + tested (`split_normal_log_kernel`, `convolve_logprob`, `ConditionalNSFFlow.log_prob_convolved`; 14/14 tests, recovers analytic Gaussian convolution).
- ■ **Clean** → **dedup** → **re-split** implemented + tested (`build_clean_manifest`; 25,200 clean rows, exact 80/10/10, leakage-safe, deterministic).
- ■ **Full wiring done + validated locally**: `prepare-data --clean` cleans/dedups/re-splits + writes extra target & error columns; `AIONHDF5Dataset` emits per-source σ (10-tuple); `main.py` `--target {…,logmstar,hr32_u} / --error-mode {none,inject,convolve} / --clean`; `batch_nll` uses `log_prob_convolved`. 14/14 tests; a real `prepare-data --clean --limit 12` produced correct columns.
- ■ **FASRC ready**: 17 GB data staged + `fits_pool` unzipped (10,355 cutouts); conda env built and verified (torch 2.6.0+cu124, aion, zuko all import); repo is a real git clone of `multitarget-clean-errors`; account `finkbeiner_lab` confirmed via `sbatch --test-only`.
- ■ **W&B tracking** wired into `train` (opt-in `--wandb`, offline-safe, no-op if absent).
- ■ Next: `prepare-data --clean` (CPU job) → V1 `log_ml_flux_1` smoke (`--error-mode convolve`, `--wandb`) on `gpu_test`.
- ■■ `siag_lab` membership pending (Coldfront) → `siag_gpu` invisible for now; using `gpu_test` (smokes) / `gpu` (real runs) under `finkbeiner_lab`.
- Known refinement: `hr32_u` piles at the arctanh clip for low-S/N HR (huge σ) — fine for flux/lx/logmstar; the HR run wants an S/N gate/clip.

10. Deferred / out of scope

SFR (needs DESI FastSpecFit VAC), Γ (=HR relabel), X-ray upper-limit censoring, full Salvato re-crossmatch of "the rest", V2 self-trained encoder, saliency/attribution.